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DATA ANALYSIS WITH PYTHON
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REPORT

TOPIC: OLIST DELIVERY TIME PREDICTION WITH LINEAR REGRESSION

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1. Introduction

1.1 Olist Context and Business Question

Olist is a Brazilian e-commerce platform headquartered in Curitiba. Olist operates as an intermediary: connecting small merchants with major e-commerce marketplaces, enabling these businesses to sell products without building their own distribution channels.

When placing an order, customers primarily care about how many days it will take for their items to arrive. Therefore, the central business question of this report is: *To what extent can we accurately estimate delivery days using only information available at the exact time of order placement?*

We chose linear regression for two main reasons. First, the target variable y (delivery days) is continuous. Second, each coefficient β explicitly shows how many days the delivery time increases or decreases when other factors remain constant. Consequently, the model offers both predictive capability and high interpretability.

1.2 Motivation: Late Delivery Leads to Low Review Scores

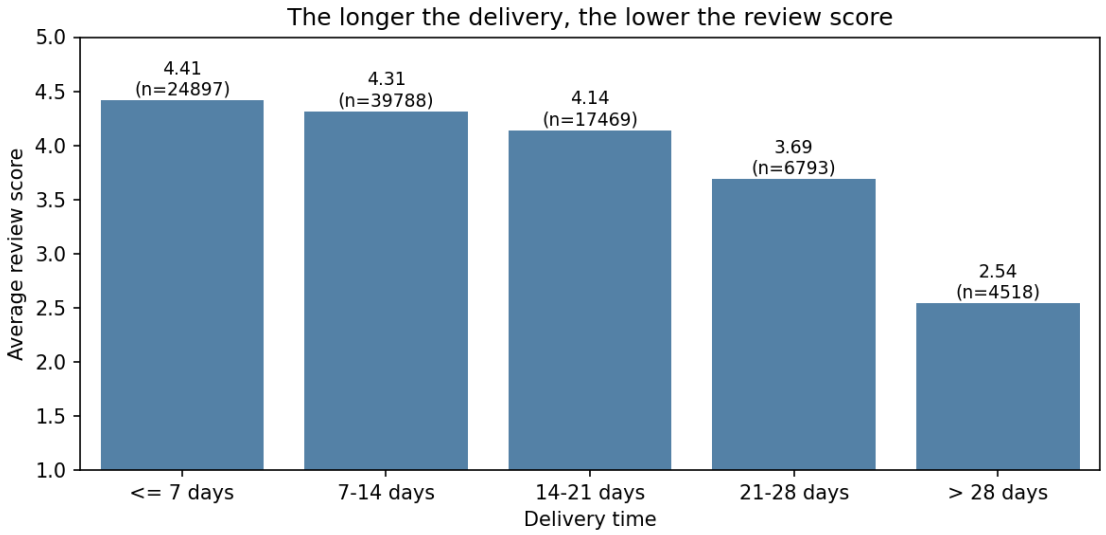


Figure 1. Average review score by delivery time interval

Interpretation. Among the 94,075 orders used in the model, 93,465 orders have reviews (99.4%). The average review score steadily declines as delivery time increases: 4.41 for orders delivered within 7 days (n = 24,897), 4.31 for the 7–14 days interval (n = 39,788), 4.14 for 14–21 days (n = 17,469), 3.69 for 21–28 days (n = 6,793), and drops to 2.54 when delivery exceeds 28 days (n = 4,518). The decrease is gradual across the first three intervals but drops sharply starting from the 21-day mark. The correlation coefficient between delivery_days and review_score is r = −0.309, indicating a moderate negative correlation: slower deliveries lead to lower customer ratings (this reflects an association, not causal proof).

Note. Reviews are submitted only after customers receive their orders; thus, they are used solely to illustrate the problem's motivation and are excluded from X.

1.3 Main Contributions, Report Structure, and Links

The main contributions of this report include:

- Building a data pipeline with row-count assertions after every join to prevent unintentional row duplication.
- Engineering geospatial features: Haversine distance between seller and customer, along with a same-state indicator flag.
- Implementing a statistical learning Pipeline (imputation, scaling, etc.) fitted exclusively on the training set to prevent data leakage.
- Benchmarking the model against two baseline models.
- Verifying linear regression assumptions and interpreting coefficients alongside p-values and confidence intervals.

Report Structure: Section 2 presents the problem definition, methodology, and evaluation; Section 3 concludes the study; Section 4 provides discussions.

Source Code: https://colab.research.google.com/drive/15G7c8a1xaL1FvOTz9hBcXeQpcmkTTJDt#scrollTo=4d6f4117
GitHub: https://github.com/tramy07/DA_Olist_LinearRegression

2. Method & Experiment

2.1 Problem Definition

2.1.1 Regression Problem Formulation

Target variable:

y = delivery_days = order_delivered_customer_date − order_purchase_timestamp (in days)

Input features (X): consists of features known at the time of purchase: seller–customer distance, weight, volume, freight value, price, item count, same-state flag, customer state, purchase month, and purchase day of the week.

Linear regression model:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$$

The objective is to find the coefficient vector β that minimizes the sum of squared errors on the training set:

$$\Sigma_i (y_i - \hat{y}_i)^2$$

2.1.2 Data Description

Data was obtained from Kaggle (Olist Brazilian E-Commerce), containing approximately 100,000 orders spanning 2016–2018, structured into several tables: orders, order_items, order_payments, order_reviews, customers, sellers, geolocation, products, and product_category_name_translation.

Table 1. Original data table dimensions (right after reading the files)

Table	Row Count	Column Count	Missing Values	Duplicate Rows
orders	99,441	8	4,908	0
order_items	112,650	7	0	0
payments	103,886	5	0	0
customers	99,441	5	0	0
products	32,951	9	2,448	0
sellers	3,095	4	0	0
geolocation	1,000,163	5	0	261,831
reviews	99,224	7	145,903	0
category_translation	71	2	0	0

Duplicate rows are removed during cleaning; for example, geolocation falls from 1,000,163 to 738,332 rows.

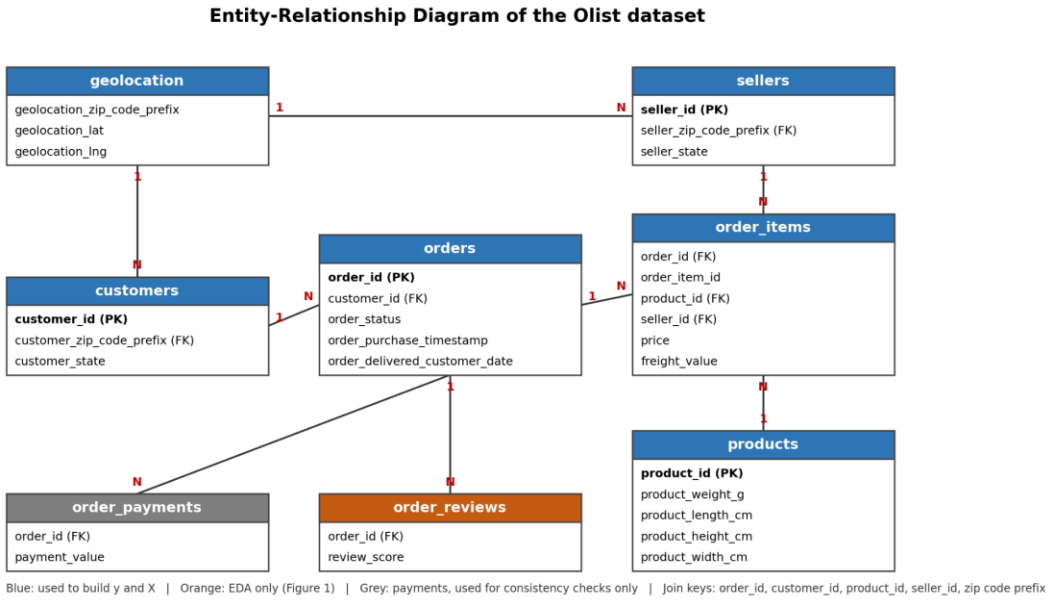


Figure 2. Entity-Relationship Diagram of the Olist dataset

Tables are linked via primary/foreign keys: order_id, customer_id, product_id, seller_id, and zip code prefix.

Note on customer identifiers: customer_id is a per-order identifier (unique per order, validated via assert), whereas customer_unique_id represents the true unique customer identifier. This study does not require customer_unique_id.

Table 2. Roles of each table

Category	Table	Notes
Used for X	orders, order_items, order_payments (validation only), customers, sellers, geolocation, products	Construct y and features
EDA only	order_reviews	Motivation illustration only (Figure 1), excluded from X
Unused	product_category_name_translation	Product category features not yet included

2.1.3 Data Leakage Prevention Principles

Strict rule: rely exclusively on information available at the exact moment of order placement.

- Exclude order_delivered_carrier_date, order_delivered_customer_date, and review data, as these are generated post-delivery.
- Exclude order_estimated_delivery_date. While Olist calculates this estimate at order placement, it is inherently an existing prediction. Including it in X would transform the objective into "re-predicting an existing prediction." This is a deliberate design choice by our team.
- total_payment is used solely for a consistency check and is excluded from X. For 99.7% of the 94,075 orders, total_payment equals total_price + total_freight within 1 BRL; the 237 exceptions (the largest gap is 182.81 BRL) are not used in X.

2.2 Method

2.2.1 Data Cleaning and Table Merging

Data cleaning steps:

1. Remove duplicate rows (none were found in orders) and cast datetime columns to datetime data types.
2. Filter for orders with delivered status only; drop orders missing actual delivery dates.
3. Aggregate order_items (total price, total freight, item count) and order_payments (total payment) to the order_id level before merging into orders. Use assert statements after each join to verify row counts and prevent unintended row duplication.
4. Drop orders without items entirely; do not impute dummy values.
5. Target variable y: remove values with $y \leq 0$, then retain values within the 1st to 99th percentiles (approximately 1.83 to 46.05 days).

Table 3. Order count statistics post-cleaning

Step	Remaining Orders	Removed Orders
Initial data (orders)	99,441	–
Keep delivered orders only	96,478	2,963
Drop delivered orders missing actual delivery date	96,470	8
Drop orders missing items (order_items)	96,470	0
Remove orders with delivery_days ≤ 0	96,470	0
Remove outliers outside 1st–99th percentiles of y	94,540	1,930
Drop orders missing distance (missing coordinates)	94,075 (final)	465

Interpretation. Following the entire data cleaning process, 94,075 orders remained, corresponding to 94.6% of the initial dataset. The step removing the highest number of orders was keeping delivered orders only (2,963 orders removed). No duplicate rows were found in the orders table; the removed orders had other statuses (shipped 1,107, canceled 625, unavailable 609, invoiced 314, processing 301, created 5, approved 2). After outlier trimming (before the 465 orders with missing coordinates were dropped), delivery_days has a mean of 12.16 days, a median of 10.22 days, and a standard deviation of 7.71 days, bounded within [1.83, 46.05] days. Model performance is evaluated strictly within this truncated distribution and does not generalize to extreme delivery durations outside this range.

2.2.2 Feature Engineering

Geospatial distance. Geolocation points outside Brazil were filtered (retaining latitudes approximately between −34 and 6, longitudes between −74 and −34). Coordinates for each zip code prefix were defined using median values. Seller–customer distance was calculated using the Haversine formula:

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2(\Delta\varphi/2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda/2)}\right), \text{ with } R = 6371 \text{ km}$$

where φ is latitude and λ is longitude (converted to radians).

Multi-seller orders. distance_km takes the maximum value, as an order is completed only when the furthest item arrives. same_state takes the minimum value (1 only if every item originates from the same state as the customer).

Weight and volume. weight_total_g and volume_total_cm3 are aggregated totals per order; item volume = length × height × width. Missing values are left as NaN and imputed with the training set median inside the Pipeline.

Freight, price, and item count. total_freight, total_price, and item_count are derived from order_items (aggregated sums or counts per order).

Categorical variables (customer_state, purchase_month, purchase_dayofweek): one-hot encoded, dropping one reference category to prevent perfect multicollinearity. Post one-hot encoding, X contains 50 columns.

Missing coordinates: orders where distance could not be computed due to missing coordinates were dropped rather than dummy-imputed; the number of orders removed at this step was 465.

Table 4. Description of input variables (X)

Variable	Data Type	Description	Source Table	Calculation Method
distance_km	Numerical	Seller–customer distance	geolocation, sellers, customers	Haversine formula; max distance for multi-seller orders
same_state	Numerical (0/1)	Seller in same state as customer	sellers, customers	1 if same state; min value for multi-seller orders
weight_total_g	Numerical	Total order weight	products, order_items	Sum aggregated by order_id
volume_total_cm3	Numerical	Total order volume	products, order_items	Length × Height × Width, sum aggregated by order_id
total_freight	Numerical	Total freight value	order_items	Sum aggregated by order_id
total_price	Numerical	Total product price	order_items	Sum aggregated by order_id
item_count	Numerical	Item count in order	order_items	Count aggregated by order_id
customer_state	Categorical	Customer state	customers	One-hot encoded, dropping one reference category
purchase_month	Categorical	Purchase month	orders	Derived from order_purchase_timestamp; one-hot encoded, dropping one level
Purchase_dayofweek	Categorical	Purchase day of week	orders	Derived from order_purchase_timestamp; one-hot encoded, dropping one level

In total, there are 7 numerical variables and 3 categorical variables; following one-hot encoding, X comprises 50 feature columns.

2.2.3 Train-Test Split

The final dataset contains 94,075 orders and 50 features. It was divided 80:20 into training and testing sets using train_test_split() with random_state = 42. The split was made before imputation, scaling and VIF, so that the test set does not influence any learned statistic.

Dataset	Number of orders	Percentage
Training set	75,260	80%
Testing set	18,815	20%
Total	94,075	100%

Before the split, observations with non-positive delivery times were removed, and delivery times outside the 1st–99th percentile range (approximately 1.83 to 46.05 days) were excluded. These percentiles were computed on the full cleaned dataset, so the test set had a minor influence on the trimming thresholds; imputation, scaling and VIF, however, use training data only. Consequently, the evaluation results describe model performance on this trimmed distribution rather than on all delivered orders, including extreme delivery times.

2.2.4 Exploratory Data Analysis (EDA)

EDA was conducted to understand the distribution of delivery time, examine correlations between numerical variables, and identify differences in delivery performance across customer states. EDA describes the whole cleaned dataset (94,075 orders), whereas VIF selection and the model use only the training set.

Distribution of Delivery Time

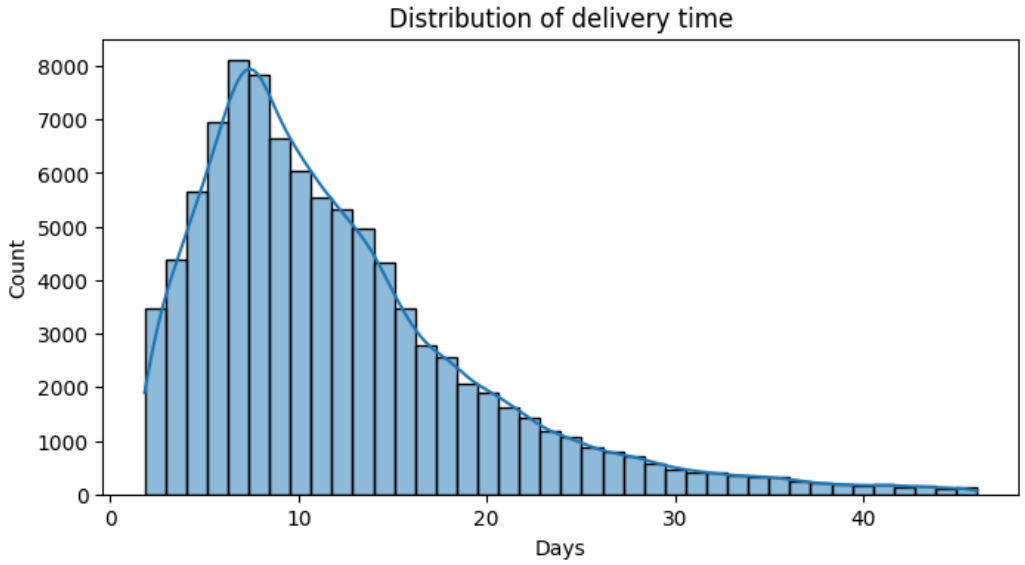


Figure 3. Distribution of delivery time

The distribution of delivery time is right-skewed: the mean (12.16 days) is higher than the median (10.22 days), and the standard deviation is 7.71 days. The 25th and 75th percentiles are 6.84 and 15.46 days. The 1%–99% trimming removes the extreme right tail (1,930 orders) but the skew remains. This is why linear regression, which assumes symmetric errors, will have difficulty with the longest deliveries (Section 2.3.3).

Correlation Analysis

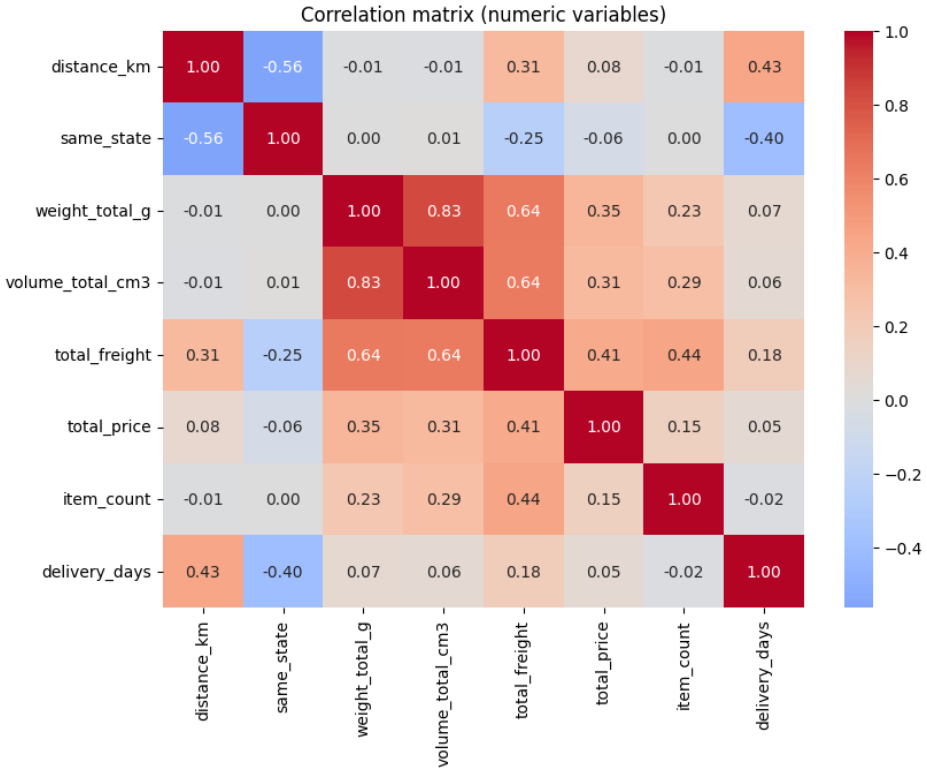


Figure 4. Correlation matrix of numerical variables

The correlation analysis shows that distance is the strongest numerical predictor of delivery time, with a positive correlation of 0.434. This indicates that orders travelling longer distances tend to take more time to arrive. In contrast, same_state has a negative correlation of −0.403 suggesting that orders whose items are all shipped from the customer's state tend to have shorter delivery times. Other numerical features, including total freight, weight, volume, price and item count, have relatively weak correlations with the target.

Among the predictors themselves, distance_km and same_state are moderately correlated (r = −0.56), weight_total_g and volume_total_cm3 are strongly correlated (r = 0.83), and total_freight is related to weight and volume (r = 0.64) and to total_price (r = 0.41). These pairs are examined in Section 2.2.5.

Delivery Time by Customer State

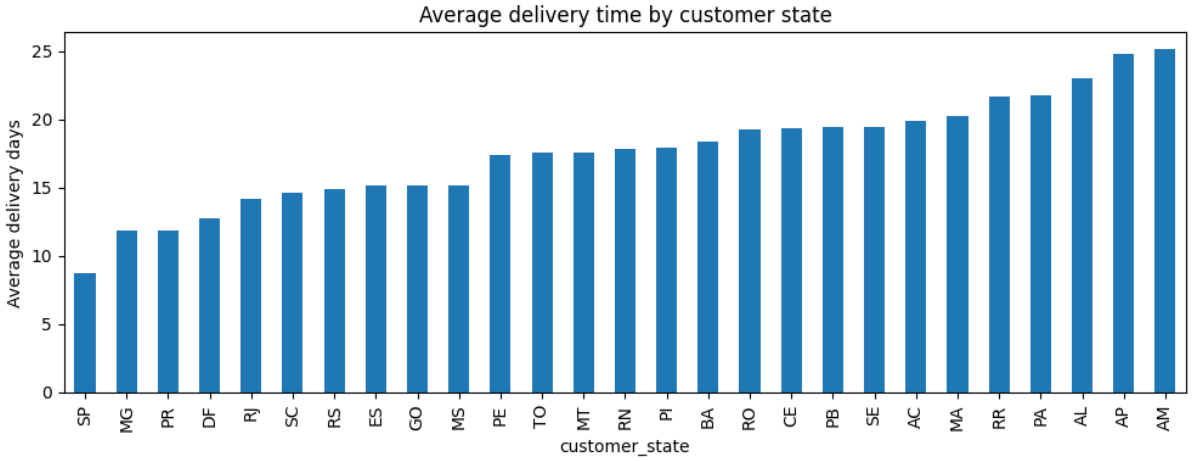


Figure 5. Average delivery time by customer state

Average delivery time differs strongly across customer states: from 8.7 days in SP and 11.8 days in MG to 23.0 days in AL, 24.8 days in AP and 25.1 days in AM, a gap of about 16 days. Distance alone may not capture this gap (for example, because of regional infrastructure), so customer state is one-hot encoded in addition to distance. One-hot encoding also avoids assuming a natural numerical order among states.

2.2.5 Feature Selection

Feature selection was conducted to examine multicollinearity among the numerical predictors before fitting the model. Pairwise correlation and Variance Inflation Factor (VIF) were calculated using the training set. Missing numerical values were first imputed with the corresponding training-set medians for this diagnostic analysis.

A VIF value above 10 is commonly treated as an indication of severe multicollinearity. In addition, pairs of variables with an absolute correlation greater than 0.7 were examined to identify potentially redundant predictors.

Table 5. Correlation with target and VIF of the numerical predictors (training set)

Numerical predictor	Correlation with delivery time	VIF	Decision
distance_km	0.434	1.657	Keep
same_state	-0.404	1.498	Keep
weight_total_g	0.071	3.616	Keep
volume_total_cm3	0.063	3.529	Keep
total_freight	0.181	2.856	Keep
total_price	0.054	1.230	Keep
item_count	-0.017	1.311	Keep

The pairwise correlation analysis identified one pair with an absolute correlation above 0.7: weight_total_g and volume_total_cm3 (r = 0.831). Although these two variables are strongly correlated, their VIF values remain below the threshold of 10. The VIF analysis was applied only to the seven numerical predictors; categorical variables were one-hot encoded separately and were not included in this VIF table.

Therefore, all seven numerical predictors were retained. Weight and volume were kept because their VIF values (3.6 and 3.5) are well below 10; however, their correlation (r = 0.83) means that their individual coefficients should be interpreted with care (Section 2.3.4). Ridge or Lasso regression, proposed in Section 4.3, would be a natural way to stabilise them.

2.2.6 Linear Regression Model

Linear Regression was selected because it is simple, fast to fit and interpretable: each coefficient shows how many days the delivery time changes when one feature changes and the others stay constant. The model is

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p + \varepsilon$$

where y is delivery time in days, β_0 is the intercept, β_j is the coefficient of feature x_j, p = 50 is the number of features and ε is the error. The coefficients are estimated by ordinary least squares (OLS):

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

which minimises the sum of squared errors on the training set.

The model assumes (i) a linear relationship between X and y, (ii) constant error variance, (iii) approximately normal errors, (iv) no severe multicollinearity and (v) independent observations. Section 2.3.3 checks these assumptions after fitting.

The model was implemented with a Scikit-learn Pipeline of three steps, all fitted on the 75,260 training observations only, so that the test set cannot influence any learned statistic:

1. **Median imputation:** Missing values (only weight_total_g and volume_total_cm3, 16 orders each) were replaced by training-set medians.
2. **Standardization:** All 50 input columns, including the one-hot dummies, were standardized with the training-set means and standard deviations.
3. **Linear Regression:** The model was fitted on the standardized features.

The pipeline was evaluated on 18,815 test observations. Because of the standardization, the fitted coefficients are per one standard deviation of each column. They were converted back to original units with

$$\beta_{original} = \beta_{std} / scale \quad and \quad \beta_{0,original} = \beta_0 - \Sigma_{j=1}^{p} (\beta_{original,j} \times mean_j)$$

For example, the coefficient of distance_km is about 0.0029 days per kilometre, and that of same_state is about −2.6593 days, meaning that orders whose items all come from the customer's state were predicted to take about 2.66 fewer days than otherwise comparable orders. The largest standardized coefficients belong to distance, same-state status, purchase month and customer state (Section 2.3.4). The coefficients represent conditional associations rather than proof of causal effects.

2.3 Evaluation & Description

2.3.1 Evaluation Metrics and Baseline Models

The model was evaluated using three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2).

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted delivery times. A lower value indicates better prediction accuracy.

$$MAE = (1/n)\ \Sigma_{i=1}^{n}\ |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error. It penalises larger errors more heavily than MAE.

$$RMSE = \sqrt{(\ (1/n)\ \Sigma_{i=1}^{n}\ (y_i - \hat{y}_i)^2\ )}$$

Coefficient of Determination (R^2) measures the proportion of variation in delivery time explained by the model. A value closer to 1 indicates a better fit, while a negative value means the model performs worse than predicting the test-set mean.

$$R^2 = 1 - \Sigma_{i=1}^{n}\ (y_i - \hat{y}_i)^2\ /\ \Sigma_{i=1}^{n}\ (y_i - \bar{y})^2$$

Two baseline models were used to assess whether Linear Regression provides an improvement over simple prediction approaches:

- **Global-mean baseline:** Predicts every test order using the average delivery time in the training set.
- **Customer-state-mean baseline:** Predicts delivery time using the average training-set delivery time for the corresponding customer state. If a state is not represented in the training set, the global training mean is used instead.

Confusion matrix and accuracy are metrics for classification problems and do not apply to a continuous target such as delivery time. They are replaced here by MAE, RMSE and R^2, together with the comparison with the two baselines.

As an additional, easily interpreted measure, we report the share of test orders whose absolute error is at most 3 or 5 days. This indicates how often the model's predictions fall within a specified number of days of the actual delivery time.

2.3.2 Model Comparison and Generalization

Performance Comparison with Baseline Models

Table 6. Performance comparison on the test set

Model	MAE (days)	RMSE (days)	R^2
Global-mean baseline	5.803	7.700	-0.000
Customer-state-mean baseline	5.062	6.855	0.207
Linear Regression	4.651	6.349	0.320

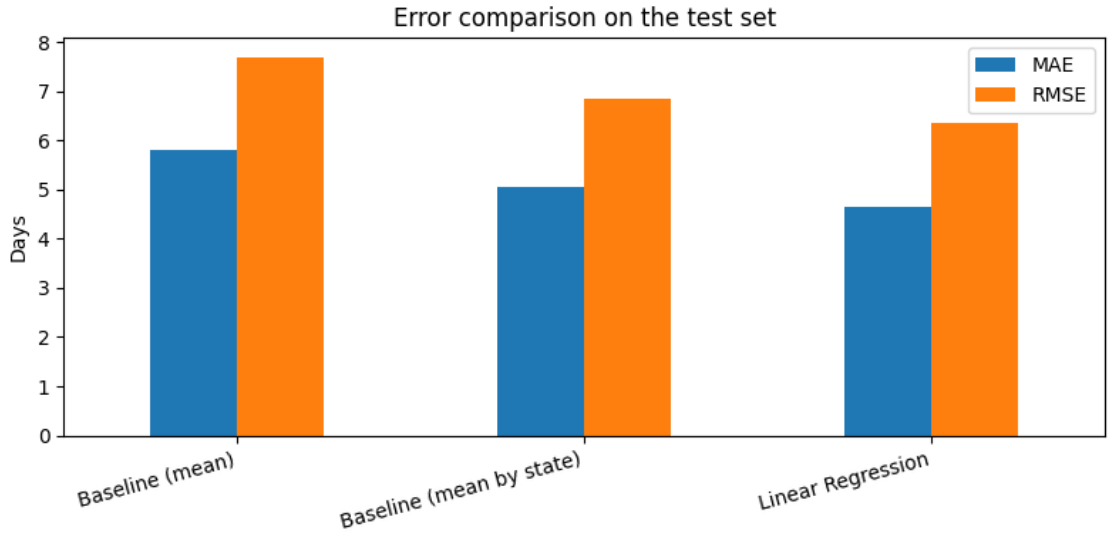


Figure 6. Model performance comparison on the test set

Linear Regression achieved the lowest prediction errors among the three approaches. Its MAE of 4.651 days is 1.15 days (19.9%) lower than the global-mean baseline and 0.41 days (8.1%) lower than the customer-state-mean baseline. Its RMSE is 17.5% lower than the global-mean baseline and 7.4% lower than the state-mean baseline.

The additional error-tolerance measure also shows that Linear Regression performs better than both baselines. For Linear Regression, 42.4% of test orders have an absolute prediction error of at most 3 days, while 65.4% have an error of at most 5 days. The corresponding proportions are 38.1% and 61.0% for the customer-state-mean baseline, and 31.1% and 52.1% for the global-mean baseline.

Model	Within 3 days	Within 5 days
Global-mean baseline	31.1%	52.1%
Customer-state-mean baseline	38.1%	61.0%
Linear Regression	42.4%	65.4%

Training and Testing Performance

Table 7. Training and testing performance

Dataset	MAE (days)	RMSE (days)	R^2
Training set	4.599	6.324	0.326
Testing set	4.651	6.349	0.320

The training and testing metrics are close (MAE 4.599 vs 4.651, a difference of 0.052 days; R^2 0.326 vs 0.320, a difference of 0.006), so there is no sign of overfitting. This does not mean the predictions are accurate: the test MAE of 4.651 days means predictions differ from actual delivery times by about 4.65 days on average.

Predicted versus Actual Delivery Time

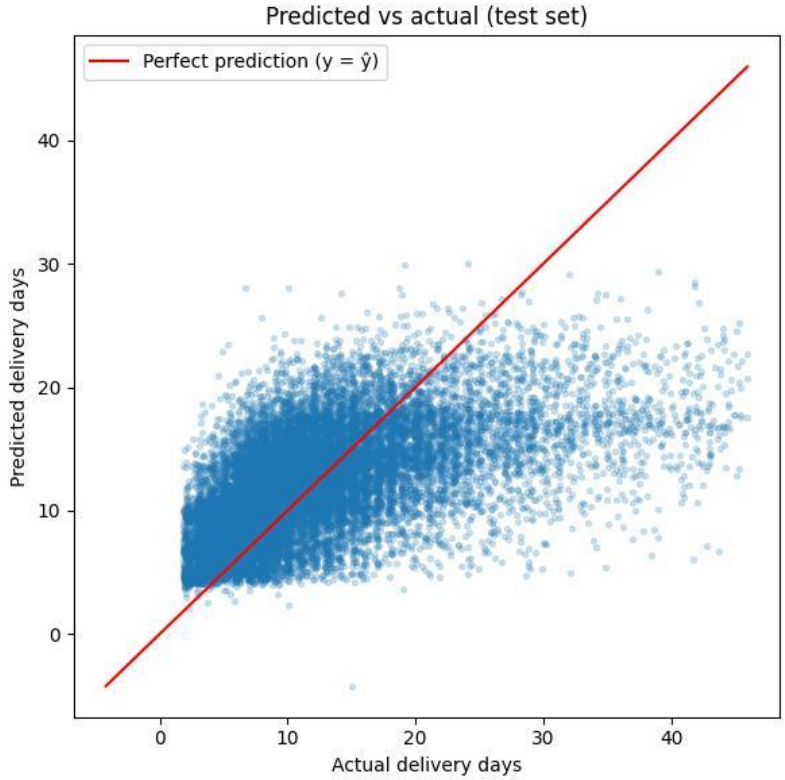


Figure 7. Predicted versus actual delivery time (test set)

The points follow the diagonal for orders delivered in about 10 to 15 days, but predictions flatten for longer deliveries. Short deliveries are overestimated: for orders that actually took 5 days or less (2,452 test orders), the mean prediction is 7.9 days against 3.6 days actual, and 98.6% of them are overestimated. Very long deliveries are underestimated: for orders that took more than 30 days (715 test orders), the mean actual time is 36.0 days but the mean prediction is only 17.4 days (MAE 18.6 days). Predictions rarely exceed 25 days (56 of 18,815 test orders). The model therefore captures the typical delivery time but not the longest delays.

2.3.3 Assumption Checking

The point predictions of a linear model remain usable even when its errors are poorly behaved, but the p-values and confidence intervals in Section 2.3.4 assume linearity, constant error variance, approximately normal errors, no severe collinearity and independent observations. We check them with the test-set residuals (actual minus predicted days). Training and testing errors are nearly identical (Section 2.3.2), so similar patterns are expected in training. The checks are visual, based on the VIF results of Section 2.2.5, and, for constant variance, a Breusch–Pagan test on the training residuals.

Residual plot

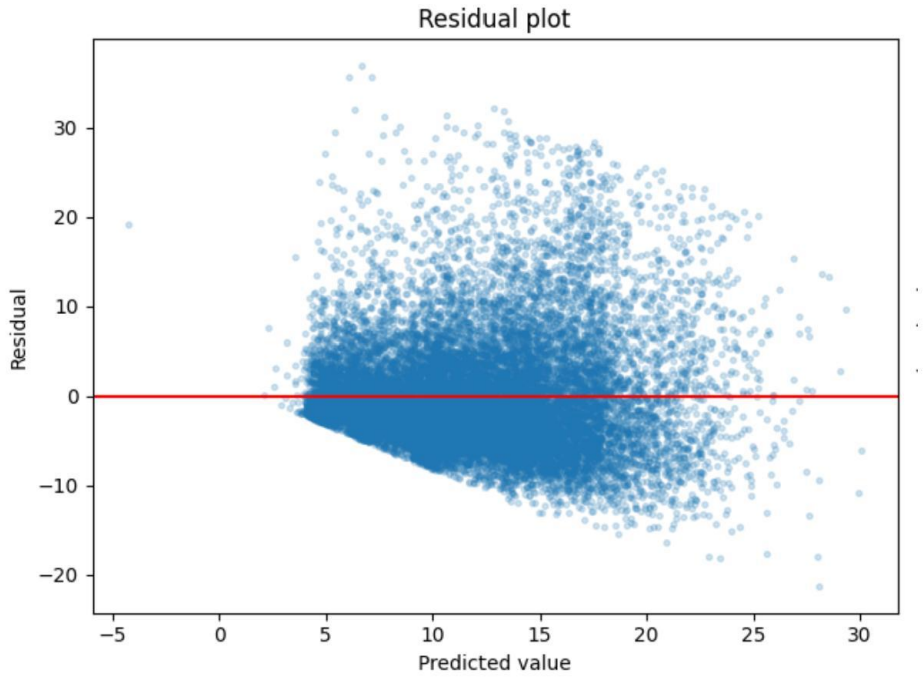


Figure 8. Residual plot (actual minus predicted delivery days) on the test set.

Figure 8 shows three features.

1. **Asymmetric errors.** About 60% of residuals are negative (median −1.2 days), but a thin tail reaches about +37 days: the model overestimates by a few days for most orders and badly underestimates a few very slow ones.
2. **A sharp lower boundary.** The straight diagonal lower edge is an artefact of trimming y at 1.83 days (the residual cannot fall below about 1.83 minus the prediction). It also means the residuals cannot be exactly normal.
3. **Widening spread.** For predictions of 5 to 8 days, 98% of residuals lie between about −5 and +19 days (standard deviation 4.7); for predictions of 12 to 18 days, they lie between about −11 and +24 days (standard deviation 7.2). The spread increases with the predicted value, which indicates heteroscedasticity. A Breusch–Pagan test on the training residuals confirms it (LM = 3668, $p < 0.001$).

No systematic curvature is visible. The predicted-versus-actual plot (Figure 7) shows that predictions rarely exceed about 25 days (56 of 18,815 test orders), and for orders that actually took more than 30 days the model typically predicts only 11 to 24 days (5th to 95th percentile). This compression is expected when R^2 is about 0.32 and is the source of the long positive residual tail.

Q-Q plot

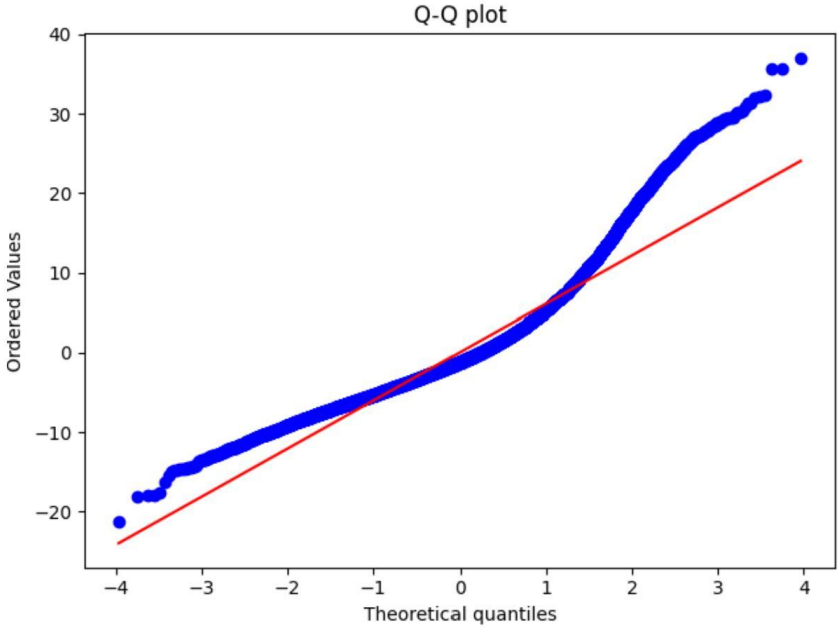


Figure 9. Q-Q plot of the test-set residuals against a normal distribution.

Figure 9 compares the residual quantiles with a normal distribution. The points follow the line between about −1 and +1, but the upper tail bends steeply above it (about +37 days against about +24 expected at quantile +4), while the lower tail is lighter than normal, consistent with the lower boundary. The residuals are right-skewed with a heavy upper tail, like the delivery times themselves (Section 2.2.4). The skewness of the residuals is 1.38 and the excess kurtosis is 2.93, confirming a right-skewed distribution with a heavy upper tail.

Summary of the assumption checks

Table 8. Assumption checks for the Linear Regression model

Assumption	Evidence	Verdict
Linearity	No clear residual curvature, but predictions flatten for longer deliveries (Fig. 7).	Partly satisfied
Constant variance	Residual spread increases with predicted values (Fig. 8; Breusch–Pagan $p < 0.001$).	Violated
Normal errors	Right-skewed residuals with a heavy upper tail (Fig. 9; skewness 1.38, excess kurtosis 2.93); lower boundary due to trimming.	Violated
Multicollinearity	All numeric VIFs below 10 (max 3.616, weight_total_g). Only weight-volume correlation exceeds 0.7 (0.831); one-hot dummies excluded.	Satisfied
Independence	Orders from the same seller or period may share unobserved shocks.	Not verified

Consequences. The violations do not affect the point predictions or the baseline comparison in Section 2.3.2, which rely only on MAE, RMSE and R^2 on unseen data. They make the standard errors behind Table 10 approximate. With heteroscedasticity-robust (HC3) standard errors, 29 of the 51 coefficients remain significant instead of 30: total_price ($p = 0.024$ with classical errors) becomes non-significant ($p = 0.053$), and total_freight remains non-significant ($p = 0.25$). Results near the threshold should therefore be treated as indicative only.

2.3.4 Coefficient Interpretation

Because the Pipeline standardises all 50 input columns, including the one-hot dummies, coefficients are given per one standard deviation (SD) of the column and converted to original units with b_original = b_std / scale and b0_original = b0 − sum(b_original × mean). Original-unit values give effects in days. For one-hot variables, the original-unit coefficient is the difference in days relative to the omitted reference level: Acre (AC), January and Monday. The standardised value of a dummy is per SD of that dummy (for example 0.31 for month_8), not per switch. The intercept (13.11 days) refers to a hypothetical order with all numeric features at zero in the reference categories and has no practical meaning.

Table 9. Top 15 coefficients ranked by absolute standardised value

Rank	Variable	b per 1 SD (days)	b in original unit (days)	Reading
1	distance_km	+1.7197	+0.0029	+0.29 days per 100 km
2	same_state	-1.2740	-2.6593	2.66 days faster when all items come from the customer's state
3	month_8	-1.1866	-3.8014	3.80 days shorter than January
4	month_7	-1.0536	-3.4382	3.44 days shorter than January
5	state_SP	-1.0265	-2.0799	2.08 days shorter than reference state AC
6	month_6	-0.9324	-3.1698	3.17 days shorter than January
7	month_5	-0.5540	-1.7960	1.80 days shorter than January
8	state_MG	-0.5212	-1.6048	1.60 days shorter than reference state AC
9	month_2	+0.4962	+1.7897	1.79 days longer than January
10	weight_total_g	+0.4387	+0.0001	about 0.09 day per kg
11	state_BA	+0.4319	+2.3719	2.37 days longer than reference state AC
12	month_9	-0.3940	-1.9308	1.93 days shorter than January
13	dow_4	+0.3896	+1.1154	Friday orders 1.12 days longer than Monday
14	month_12	+0.3604	+1.5598	1.56 days longer than January
15	month_4	-0.3593	-1.2286	1.23 days shorter than January

Variables not listed (volume_total_cm3, total_freight, total_price, item_count and the remaining states, months and weekdays) all have an absolute standardised coefficient of at most 0.36 days per SD. For dummy variables, "b per 1 SD" refers to the SD of the dummy; the original-unit column is the effect per switch from the reference level. Ranking by original-unit effect would place the regional dummies state_AM (+5.58 days), state_AL (+5.35) and state_AP (+4.69) ahead of same_state; they rank lower here only because these states represent a very small share of orders and so have a small SD.

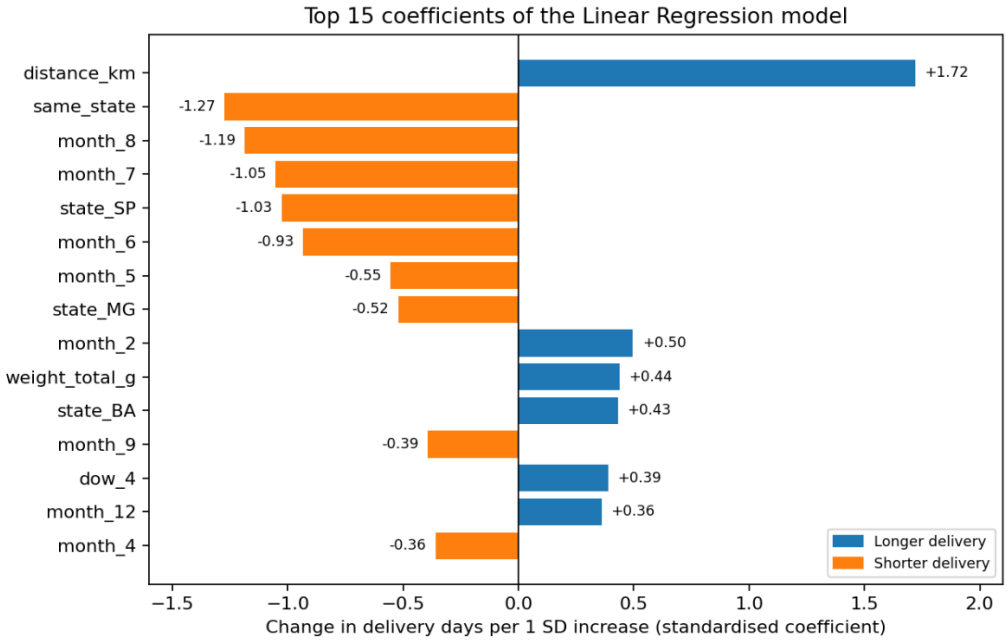


Figure 10. Top 15 coefficients by absolute standardised value (change in delivery days per one standard deviation of each column; for dummy variables, per SD of the dummy rather than per switch from the reference level).

Statistical significance

An OLS model (statsmodels) fitted on the training set in original units gives the p-values and confidence intervals in Table 10 (intercept and numeric predictors; training $R^2 = 0.326$, the same as the Pipeline). Of the 51 parameters (50 features plus the intercept), 30 are significant at the 5% level; among the numeric predictors only total_freight is not.

Table 10. OLS coefficients of the intercept and numeric predictors (training set)

Variable	b	95% CI lower	95% CI upper	p-value	Significant (p < 0.05)
Intercept	13.1114	11.3519	14.8709	< 0.0001	Yes
distance_km	0.0029	0.0027	0.0031	< 0.0001	Yes
same_state	-2.6593	-2.8127	-2.5058	< 0.0001	Yes
weight_total_g	0.0001	0.0001	0.0001	< 0.0001	Yes
volume_total_cm3	0.0000	0.0000	0.0000	< 0.0001	Yes
total_freight	-0.0031	-0.0067	0.0005	0.0929	No
total_price	-0.0003	-0.0005	-0.0000	0.0237	Yes
item_count	-0.5013	-0.5976	-0.4049	< 0.0001	Yes

Values are rounded to four decimals, so weight_total_g and volume_total_cm3 appear with little precision; p-values shown as 0.0000 in the notebook are reported as below 0.0001.

Reading the numeric coefficients

- **Distance.** Each additional 100 km adds about 0.29 days (95% CI 0.27 to 0.31); across the interquartile range of distances (about 199 to 800 km) this is about 1.7 days. Distance has the largest standardised effect (+1.72 days per SD).
- **Same state.** Orders whose items all come from the customer's state arrive 2.66 days sooner (95% CI 2.51 to 2.81), beyond the effect of shorter distance (the two are correlated, r = -0.56). An interstate order over 2,000 km is predicted to take about 8 days longer than a same-state order over 50 km, other features equal (0.0029 × 1,950 + 2.66).
- **Item count.** Each extra item is associated with 0.50 fewer days (95% CI 0.40 to 0.60). The sign is counter-intuitive: the marginal correlation is almost zero (r = -0.018), and the negative sign appears only after controlling for weight, volume and freight. It should not be read as advice to add items to an order.
- **Other numeric variables.** total_freight is not significant (p = 0.093) because distance and weight largely explain it (r = 0.31 and 0.64). total_price is significant with classical errors (p = 0.024) but negligible (about -0.3 days per 1,000 BRL), and it is no longer significant with robust errors (p = 0.053). Weight adds about 0.09 day per kg, and volume has a standardised effect below 0.36 days per SD; the two are strongly correlated (r = 0.831), so their individual signs need care.

Reading the categorical coefficients

- **Month.** Relative to January, orders placed from May to September are 1.8 to 3.8 days faster, and orders placed in February and December are 1.6 to 1.8 days slower. These coefficients mix seasonality, events of 2016 to 2018 (the notebook mentions Black Friday 2017 and the 2018 truckers' strike) and uneven coverage of months across years, because month is not separated from year. No coefficient is therefore attributed to a single event.
- **Customer state.** Relative to Acre (a remote, slow state), customers in SP and MG receive orders 2.08 and 1.60 days sooner, whereas customers in BA, AL, AM and AP receive them 2.37, 5.35, 5.58 and 4.69 days later, after controlling for distance and same_state. These are comparisons with a slow reference, not absolute times, and the larger effects belong to states with very few orders.
- **Day of week.** Friday orders are about 1.12 days slower than Monday orders, consistent with weekend pauses in processing and dispatch (not confirmed by the data).

Association, not causation. All coefficients describe conditional associations in observational data. They indicate which factors move together with delivery time, not what would happen if a factor were changed.

3. Conclusion

3.1 Answer to the Business Question

The central question was how accurately delivery days can be estimated from information available at order placement. On the 18,815 test orders, the model achieves an MAE of 4.651 days, an RMSE of 6.349 days and an R^2 of 0.320. The typical absolute error is about 4.7 days, roughly 38% of the average delivery time (12.16 days).

The model beats both baselines: its MAE is 19.9% lower than the global-mean baseline (5.803 days) and 8.1% lower than the state-mean baseline (5.062 days), its RMSE is 17.5% and 7.4% lower, and its R^2 rises from 0.207 to 0.320. Training and testing metrics are nearly identical (MAE 4.599 versus 4.651; R^2 0.326 versus 0.320), so there is no overfitting. The limit is that about 68% ($1 - R^2$) of the variation is not explained by checkout-time features.

Checkout-time information therefore gives a useful, better-than-average estimate but not a precise delivery promise, within the trimmed range of 1.83 to 46.05 days.

3.2 Key Drivers, Business Value and Limits

Strongest drivers. The most influential factors are, in order of standardised effect (with the caveat on dummy variables in Section 2.3.4):

1. Seller-customer distance (+0.29 days per 100 km; +1.72 days per SD).
2. Same-state shipping (-2.66 days; -1.27 days per SD).
3. Calendar and regional effects: purchase month (up to 3.8 days shorter than January in August) and customer state (SP and MG faster, BA, AL, AM and AP slower than the reference AC). Ranked by original-unit effect these regional and monthly dummies can exceed the same-state effect, but they reflect seasonality and uneven coverage across years rather than a controllable factor.

Product-related variables (weight, volume, freight, price, item count) have much smaller effects, and weight is the only one among them that reaches the top 15.

Business value. The model can show an estimated delivery time at checkout using only data available at that moment, and its coefficients are readable by non-technical teams (for example, orders whose items all come from the customer's state are predicted to arrive about 2.7 days sooner). Since review scores fall from 4.41 within 7 days to 2.54 beyond 28 days (Section 1.2), more realistic estimates can help manage customer expectations.

Limits. Two thirds of the variation is unexplained, and the longest deliveries, which matter most for reviews, are systematically underestimated (Figure 7). The output should therefore be shown as an expected value or range, not a guaranteed date.

4. Discussion

4.1 Limitations

Data and features.

- Haversine is a straight-line distance and coordinates are zip-prefix medians, so both are approximate; 465 orders with missing coordinates were dropped.
- Key drivers are missing: seller processing time, carrier, product category (the translation table was unused), holidays and weather. The estimated delivery date was excluded deliberately, so the study does not show whether the model beats Olist's own estimate.
- The data start in October 2016 and end in August 2018, and 2016 contributes only 258 orders. Months September to December are therefore observed almost only in 2017, whereas January to August are observed in both 2017 and 2018. Month effects mix seasonality, year and events.

Model.

- The target is right-skewed and the model is additive and linear, with no interactions such as distance × same_state; it can even predict impossible values (one test prediction is about -4 days).
- Heteroscedastic (Breusch–Pagan $p < 0.001$), non-normal residuals make p-values and confidence intervals approximate, and independence between orders could not be verified. With robust (HC3) errors, total_price is no longer significant.
- Some coefficients (the sign of item_count, the month effects) cannot be interpreted causally.

Evaluation.

- The 80/20 split is random, so training and test orders share the same months and events; performance on future periods may be lower.
- The 1st and 99th percentile thresholds for trimming y were computed on the full dataset before the train-test split, so the test set had a minor influence on them.

4.2 Scope of Applicability

- Results apply only to the trimmed distribution (1.83 to 46.05 days) and to delivered orders; extremely late orders were not evaluated, so the model should not be used to judge them.
- The data cover one Brazilian marketplace in 2016 to 2018; logistics and customer expectations may have changed.
- Coefficients are associations, not causal effects.

4.3 Directions for Improvement

Apart from the robust standard-error check reported in Section 2.3.3, none of the following proposals was run, so no performance gain is claimed.

1. **Transform the target.** log(delivery days) or Gamma regression to reduce skew and avoid negative predictions.
2. **Add interactions.** For example distance × same_state, or state-specific distance slopes.
3. **Regularise.** Ridge or Lasso to stabilise the correlated weight and volume coefficients.
4. **Use non-linear models.** Random Forest or Gradient Boosting on the same split.
5. **Enrich features.** Seller historical processing speed (from the training set only), product category, holidays, a time index and road distance.
6. **Strengthen evaluation.** Chronological split, prediction intervals (for example quantile regression) and a comparison with Olist's own estimate.

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